

ECO 362: Financial Investments

Options

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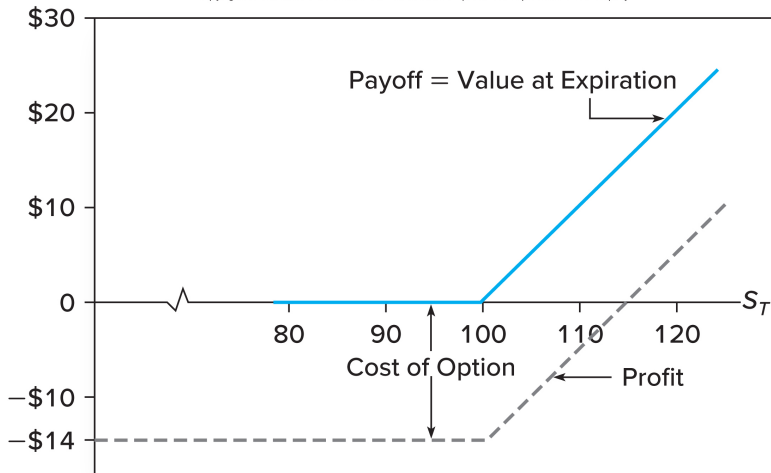
Princeton University

Call option

- Buyer pays seller option price C today.
- Buyer has right to buy an asset at exercise price X on expiration date.
- T years to expiration date.
- If the asset price is S_T on expiration date, payoff is
 - Buyer: $\max(S_T - X, 0)$
 - Seller: $-\max(S_T - X, 0)$

Payoff of buying a call option

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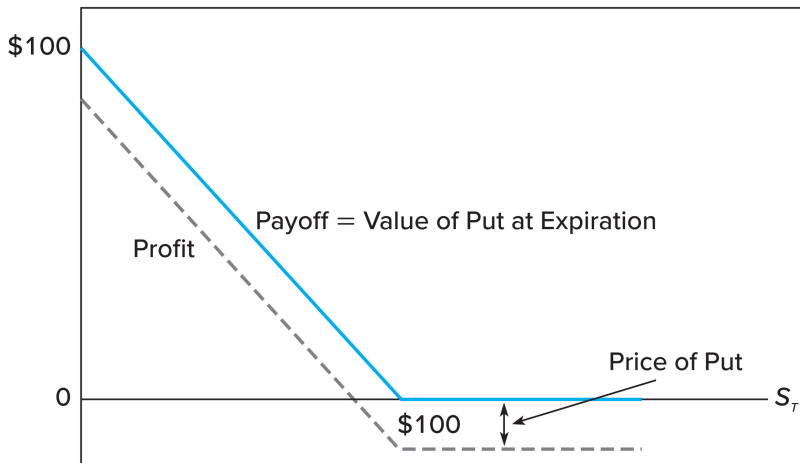
Source: Bodie, Kane, and Marcus (2018, Figure 20.2)

Put option

- Buyer pays seller option price P today.
- Buyer has right to sell an asset at exercise price X on expiration date.
- T years to expiration date.
- If the asset price is S_T on expiration date, payoff is
 - Buyer: $\max(X - S_T, 0)$
 - Seller: $-\max(X - S_T, 0)$

Payoff of buying a put option

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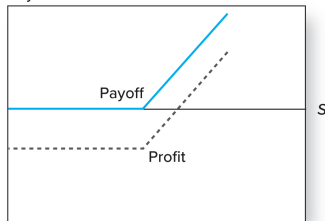


Source: Bodie, Kane, and Marcus (2018, Figure 20.4)

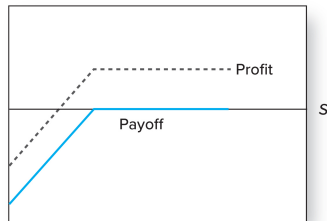
Summary of option payoffs

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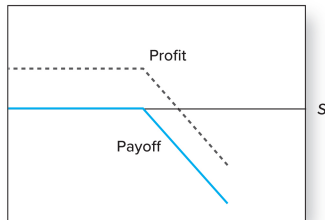
Buy Call



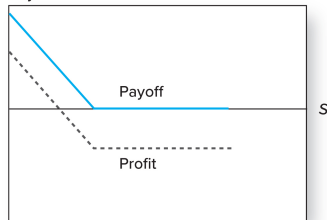
Write Put



Write Call



Buy Put



Source: Bodie, Kane, and Marcus (2018, Figure 20.3)

Types of options

- Stock options: IBM stock.
- Index options: S&P 500 index.
- Futures options: S&P 500 futures.
- Currency options: Dollar/pound exchange rate.
- Commodity options: Crude oil.

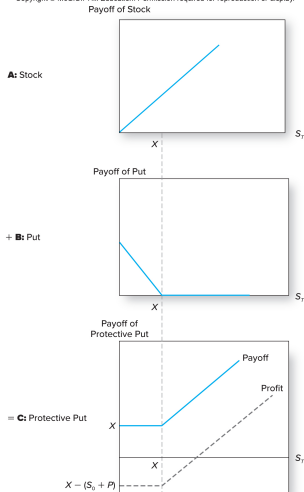
Protective put

- Protective put strategy.
 1. Buy the asset at price S_0 .
 2. Pay P for a put with exercise price X and time to expiration T .
- Payoff at expiration date.
 1. Asset worth S_T .
 2. Put payoff is $\max(X - S_T, 0)$.
- Total payoff is

$$S_T + \max(X - S_T, 0) = \max(X, S_T)$$

Payoff of a protective put

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Source: Bodie, Kane, and Marcus (2018, Figure 20.6)

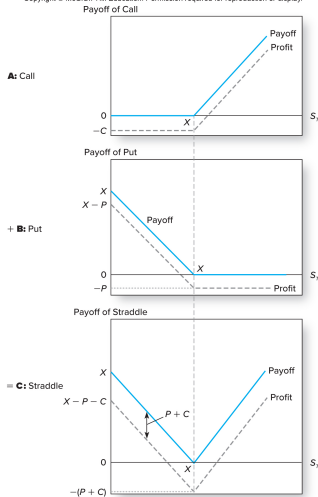
Straddle

- Straddle strategy.
 1. Pay C for a call with exercise price X and time to expiration T .
 2. Pay P for a put with exercise price X and time to expiration T .
- Payoff at expiration date.
 1. Call payoff is $\max(S_T - X, 0)$.
 2. Put payoff is $\max(X - S_T, 0)$.
- Total payoff is

$$\max(S_T - X, 0) + \max(X - S_T, 0)$$

Payoff of a straddle

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Source: Bodie, Kane, and Marcus (2018, Figure 20.9)

Summary of option strategies

- Protective put.
 - Upside potential and downside protection.
- Straddle.
 - Long: Bet on high volatility.
 - Short: Bet on low volatility.

Put-call parity

- Investment strategy.
 1. Pay P for a put with exercise price X and time to expiration T .
 2. Buy the asset at price S_0 .
 3. Borrow $\exp(-r_b T)X$.

- Total cost is

$$P + S_0 - \exp(-r_b T)X$$

- Payoff at expiration date.
 1. Put payoff is $\max(X - S_T, 0)$.
 2. Asset worth S_T .
 3. Repay X .

- Total payoff is

$$\max(X - S_T, 0) + S_T - X = \max(S_T - X, 0)$$

- Same as a call option!

- By no arbitrage, call price is

$$C = P + \underbrace{S_0 - \exp(-r_b T)X}_{\text{levered long}}$$

- Equivalently, put price is

$$P = C - \underbrace{S_0 + \exp(-r_b T)X}_{\text{levered short}}$$

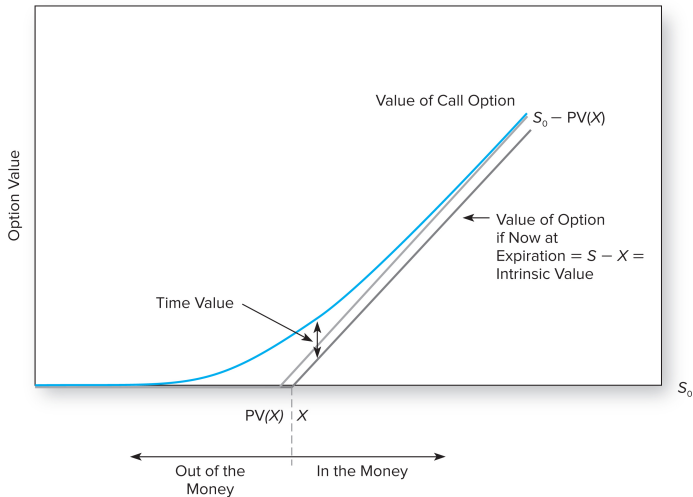
- Options are risky because of embedded leverage.

Moneyness of options

- In the money at $t \leq T$ if
 - Call: $S_t > X$.
 - Put: $S_t < X$.
- Out of the money at $t \leq T$ if
 - Call: $S_t < X$.
 - Put: $S_t > X$.
- $\max(S_t - X, 0)$ is the intrinsic value of a call option.
- Call option is worth more than the intrinsic value because of the “time value of money”, which increases in volatility.

Call option value

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Source: Bodie, Kane, and Marcus (2018, Figure 21.1)

Pricing a call option by no arbitrage

- Let M_T be the stochastic discount factor for discounting a cash flow in T years.
- Call price is

$$C = \mathbb{E}[M_T \max(S_T - X, 0)]$$

- Black-Scholes assumptions.
 - Continuously compounded riskless interest rate r_b is constant.
 - Asset returns are normally distributed with constant standard deviation (a.k.a. volatility) σ .
 - Asset does not pay a dividend prior to expiration date.

Black-Scholes model

- Call price is

$$C = S_0 \times \mathbb{N}(d_1) - \exp(-r_b T) X \times \mathbb{N}(d_2)$$

where

$$d_1 = \frac{\ln(S_0/X) + (r_b + \sigma^2/2)T}{\sigma\sqrt{T}}$$

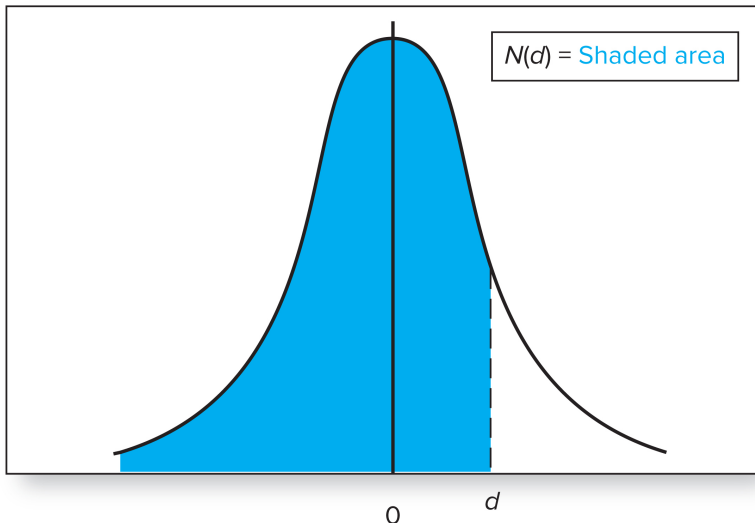
$$d_2 = d_1 - \sigma\sqrt{T}$$

- $\mathbb{N}(d)$ is probability that a standard normal random variable is less than d .
- By the put-call parity,

$$\begin{aligned} P &= C - S_0 + \exp(-r_b T) X \\ &= -S_0(1 - \mathbb{N}(d_1)) + \exp(-r_b T) X(1 - \mathbb{N}(d_2)) \end{aligned}$$

Normal distribution

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Source: Bodie, Kane, and Marcus (2018, Figure 21.6)

IBM option prices

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PRICES AT CLOSE JUNE 30, 2016

I B M (IBM)

Underlying Stock Price: 149.60

Expiration	Strike	Call			Put		
		Last	Volume	Open Interest	Last	Volume	Open Interest
July 15, 2016	145	5.18	37	6319	0.48	79	5659
August 19, 2016	145	6.98	137	1943	3.60	512	2476
October 21, 2016	145	8.42	12	1310	6.35	80	1086
July 15, 2016	150	1.85	478	6521	1.81	1649	5763
August 19, 2016	150	4.10	472	2373	5.91	81	2911
October 21, 2016	150	5.43	23	2604	8.42	77	3101
July 15, 2016	155	0.79	337	13492	5.95	77	3847
August 19, 2016	155	1.90	113	5215	8.45	9	1748
October 21, 2016	155	3.60	8	1681	11.70	12	728

Source: Bodie, Kane, and Marcus (2018, Figure 20.1)

Example 1: IBM option on 6/30/2016

- Riskless rate: $r_b = 0.20\%$
- Stock price: $S_0 = \$149.60$
- Volatility: $\sigma = 21\%$
- Exercise price: $X = \$155$
- Expiration date: 7/15/2016
- Time to expiration (years): $T = 0.04$

Option prices implied by the Black-Scholes model

- First compute

$$d_1 = \frac{\ln(149.60/155) + (0.002 + 0.214^2/2)0.04}{0.214\sqrt{0.04}} = -0.79$$

$$d_2 = d_1 - 0.214\sqrt{0.04} = -0.83$$

- Call price is

$$\begin{aligned} C &= 149.60 \times \mathbb{N}(d_1) - \exp(-0.002 \times 0.04)155 \times \mathbb{N}(d_2) \\ &= 0.79 \end{aligned}$$

- Put price is

$$\begin{aligned} P &= -149.60(1 - \mathbb{N}(d_1)) + \exp(-0.002 \times 0.04)155(1 - \mathbb{N}(d_2)) \\ &= 6.17 \end{aligned}$$

- Compare with market prices: $C = \$0.79$ and $P = \$5.95$.

Implied volatility

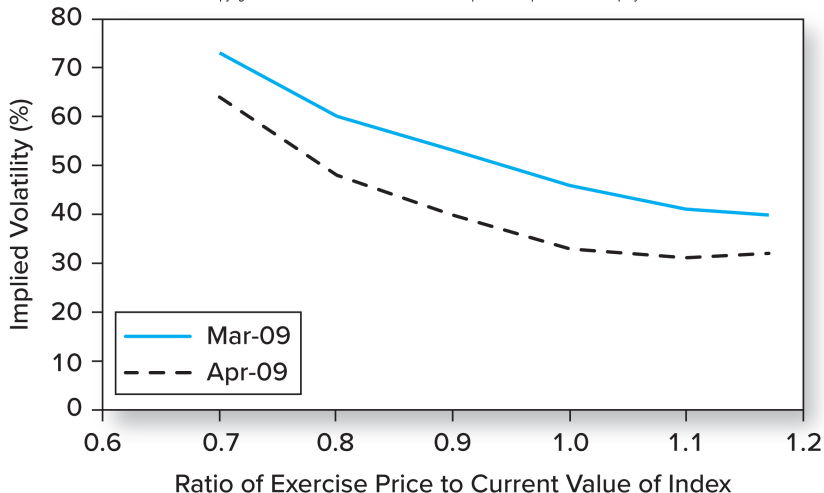
- Instead of using the Black-Scholes model to price an option for a given σ , reverse the question.
- What value of σ (called implied volatility) is consistent with market price?
- Two reasons to do this.
 1. Test the Black-Scholes model, which implies that all options with the same expiration date have the same implied volatility.
 2. Traders use implied volatility to quote option prices. Higher implied volatility means higher option price.

Example 2: IBM option on 6/30/2016

- Riskless rate: $r_b = 0.20\%$
- Stock price: $S_0 = \$149.60$
- Exercise price: $X = \$155$
- Expiration date: 7/15/2016
- Time to expiration (years): $T = 0.04$
- Use Excel solver to compute implied volatility.
 - Call price $\$0.79 \Rightarrow \sigma = 21.45\%$
 - Put price $\$5.95 \Rightarrow \sigma = 18.79\%$

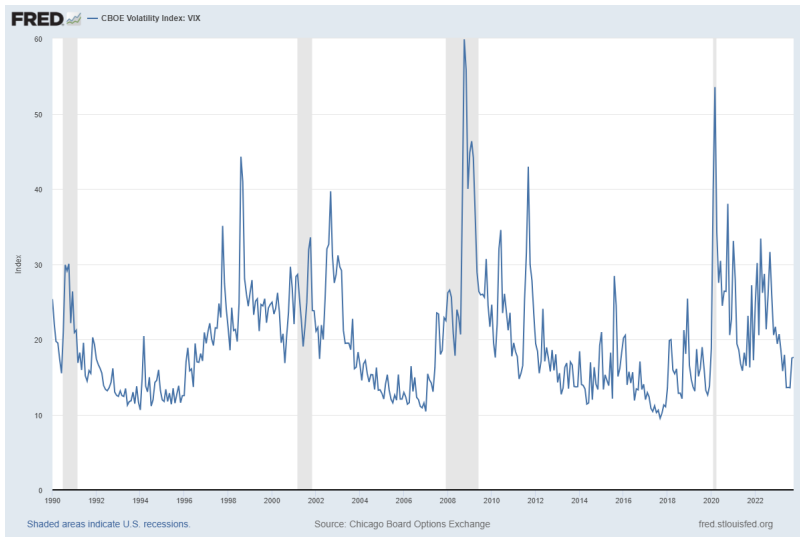
Volatility smile

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Source: Bodie, Kane, and Marcus (2018, Figure 21.15)

VIX: Implied volatility of S&P 500 index options



Source: FRED Economic Data (9/30/2023)

Summary

- Black-Scholes was the first option pricing model.
- More recent models allow for
 - Time-varying interest rates.
 - Time-varying volatility.
 - Pricing American and other exotic options.
- Covered in ORF 335/ECO 364 (Introduction to Financial Mathematics).